A Pre-Registered Cost Model for a Small Bit-Exact int8 CNN Accelerator on a RISC-V Microcontroller SoC

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Abstract—We present mcuaccel, an int8 convolution, depthwise and fully connected accelerator memory-mapped into a PicoRV32 SoC, together with a backend for mcugrad that emits accelerator command descriptors and driver C while keeping every other operator in the original software kernels. The accelerator targets bit-exactness with the TFLite-style integer reference. Its cycle counts are predicted by an analytic port-bound model whose predictions were committed to git (tag m1, 6533b1e) before any datapath RTL existed. At $P = 8$ the model predicts a speedup of 27.0× on mnist_cnn and 363.1× on cifar_mobilenet over software baselines of 12,188,028 and 72,157,425 cycles per inference. Measured: [MEASURE: speedup mnist cnn P=8]× and [MEASURE: speedup cifar mobilenet P=8]×. Claim C1 is [MEASURE: C1 verdict] and C4a is [MEASURE: C4a verdict]. All cycle counts are Verilator cycles, not FPGA wall-clock.

I. Introduction

Quantized CNNs cost tens of millions of cycles per image on microcontroller-class cores. A dedicated int8 accelerator beating a scalar in-order RISC-V core by an order of magnitude or more is well established [1]–[3]; reproducing that is not our contribution. We ask instead how well a simple analytic model, written down before the RTL, predicts the cycle counts of a small memory-bound accelerator, with tolerances fixed in advance. The claims below are pre-registered in docs/SPEC.md; the predictions are frozen at tag m1 (commit 6533b1e, m1_predictions.csv sha256 prefix fd94f39a1a913ee6).

- C1 Speedup at $P = 8$ is at least 10× on both networks, end to end with fallback layers and driver. C1b: at least 100× on cifar_mobilenet.
- C2 Bit-exactness: zero mismatching bytes on 1,000 random layers (C2a), in every intermediate tensor of both networks (C2b), and in final outputs for 100 images per network on the full SoC (C2c).
- C3 At $P = 8$ the accelerator uses at most 6,000 LUT4, 40 MULT18X18D and 12 DP16KD on an ECP5 LFE5U-85F.
- C4 The frozen model predicts every accelerator command within 20 % (C4a, all P) and each network's accelerator-plus-driver time within 10 % (C4b).
- C5 The saturation width of the P sweep is reported, predicted versus measured; its mechanism was stated in advance.

PicoRV32 → adapter → interconnect → {RAM port A, console, accelerator registers}
fetch (only master) → ctrl (descriptor FSM, loop nest) → align → MAC array (P outputs × 4 lanes) → requant → writer → RAM port B

Fig. 1. System block diagram (schematic; docs/DESIGN.md Sec. 1).

II. Background and Related Work

Quantization. The numerical contract is the integer-only scheme of Jacob et al. [4] as implemented in gemmlowp [5] and specified by LiteRT [6]: a saturating rounding doubling high multiply followed by a rounding divide by a power of two. MCU inference. TFLite Micro [7], CMSIS-NN [8] and MCUNet [9] provide software kernels; mcugrad [10] is the compiler whose int8 simulator is our specification. Accelerators. Eyeriss [11], the TPU [12], NVDLA [13], VTA [3], Gemmini [1] and CFU Playground [2] show large speedups over in-order cores; FINN [14] and PULP-NN [15] target FPGA and multicore MCU settings. None of their RTL is used here. Models. Earlier analytic models [16], [17] and the roofline model [18] were fitted alongside their designs; ours was frozen first. The CPU is PicoRV32 [19] on a Lattice ECP5 [20], with the bus protocol of corv32 [21].

III. Architecture

The CPU reaches a dual-port RAM through an adapter and interconnect; the accelerator is a second bus master on the RAM's other port (a private 32-bit port, one word per cycle, one outstanding request), so no arbiter is needed. The CPU writes a descriptor address and a start bit, then polls a status register (Fig. 1).

Dataflow. The array is output-stationary over one output pixel and a tile of P output channels. In NHWC layout the in-bounds part of a kernel row is contiguous, so each cycle one four-byte activation word meets one weight-buffer row of $4{\cdot}P$ bytes and performs $4{\cdot}P$ multiply-accumulates; every channel count uses one mechanism. Depthwise layers get only 4 useful MACs per cycle. Weights and per-channel parameters are loaded once per command into local buffers (512 rows). A single requantizer emits one byte per cycle using the specification's SRDHM/RDBPOT rule; accumulators are 32 bit and the backend enforces a no-wrap bound.

Interface. A command is a 18-word descriptor in RAM (geometry, strides, addresses, zero points, clamp). Descriptors chain by pointer and the accelerator writes a cycle stamp into each. Anything outside the conv/depthwise/fully-connected envelope falls back to software.

IV. Backend and Driver

The backend (backend/mcuaccel_backend) consumes mcugrad's int8 graph, packs weights and parameters, splits layers that exceed buffer limits, builds descriptors, and emits a model.c with the same model_run API; unsupported ops are the identical mcugrad kernels. A pure-Python executor of the descriptor semantics, tested equal to the integer reference, is the oracle for the RTL's request stream. The driver writes the descriptor address, starts the chain, polls, and patches only the fields that depend on the network input and output. Its cost, measured on the walking-skeleton stub, is 71 cycles per call plus 15 per patched field.

V. Methodology

Pre-registration. Tag m1 precedes every commit that adds a datapath or sequencer. The model is not changed to match the RTL, nor the RTL tuned to the model: the RTL is scored at its first full measurement and later cycle-changing edits are logged with before/after values. Predicted rows are those of m1_predictions.csv with architecture os (17 commands at $P = 8$).

Timing protocol. Inference time is the rdcycle interval around model_run on the SoC, including software layers, descriptor patching, polling and driver cost, over 10 fixed test images and 3 timed runs after a warm-up (decision D17: the simulator is deterministic, so more runs add nothing). We report the median of per-run per-image medians with min and max. The bench window costs 120 cycles in both builds. Cycle counts come from Verilator simulation of the full SoC at a nominal 50 MHz.

Fairness. The baseline is mcugrad's generated model.c, unchanged, built with the same compiler and flags on the same SoC with the accelerator present and idle. The accelerated build contains the same fallback kernels, checked by comparing per-op profile cycles of the two builds. Optional backend passes are off in every scored build.

VI. Results

Baseline. Table I gives the software baseline. Convolution and fully connected layers account for 96.4 % (mnist) and 99.9 % (cifar) of baseline cycles in the profile build, which bounds the Amdahl ceiling.

C1/C1b: speedup. Table II compares the frozen prediction with measurement. At $P = 8$ the prediction puts the software ops at 95.8 % (mnist) and 31.6 % (cifar) of the accelerated run: pooling and layout ops that stay in software dominate MNIST, whose predicted speedup saturates near 27.2× at the largest P. C1 is [MEASURE: C1 verdict]; C1b is [MEASURE: C1b verdict].

TABLE I
Software baseline on the SoC (cycles per inference).

Network	median	min	max
mnist_cnn	12,188,028	12,183,011	12,193,076
cifar_mobilenet	72,157,425	72,132,576	72,174,963

TABLE II
Speedup versus P: frozen prediction and measurement.

	mnist		cifar	
P	pred.	meas.	pred.	meas.
1	22.9	?	111.7	?
2	25.2	?	187.1	?
4	26.4	?	282.0	?
8	27.0	?	363.1	?
16	27.2	?	411.4	?

C2: bit-exactness. Random layers: [MEASURE: C2a layers run] run, [MEASURE: C2a mismatching bytes] mismatching bytes (C2a [MEASURE: C2a verdict]). Intermediate tensors: [MEASURE: c2b images] images, [MEASURE: c2b mismatch] mismatches. Full SoC: [MEASURE: c2c images] images, [MEASURE: c2c mismatch] mismatches.

C3: resources. Table III reports the Yosys/nextpnr-ecp5 flow on the LFE5U-85F. The accelerator at $P = 8$ uses [MEASURE: accel lut4] LUT4, [MEASURE: accel mult18x18d] MULT18X18D and [MEASURE: accel dp16kd] DP16KD; C3 is [MEASURE: C3 verdict].

C4/C5: prediction accuracy. Table IV lists predicted command cycles at $P = 8$ against the stamp measurement. Over the commands of the sweep the maximum absolute error is [MEASURE: C4a max error] %, the median signed error [MEASURE: C4a median signed error] %, with [MEASURE: C4a violations] tolerance violations out of [MEASURE: C4a rows] rows (C4a [MEASURE: C4a verdict]). Per network, accelerator plus driver, the signed error is [MEASURE: C4b error mnist cnn] % (mnist) and [MEASURE: C4b error cifar mobilenet] % (cifar). Figure 2 shows predicted per-command cycles versus P. The predicted saturation width is 16 (mnist) and 16 (cifar) on the grid, where 16 means not reached, against measured [MEASURE: measured saturation width] and [MEASURE: measured saturation width]. The model attributes the flattening to the depthwise layers, the single requantizer and the weight load, not to the port.

VII. Discussion and Limitations

All cycle counts come from Verilator simulation; the FPGA is used only for synthesis figures (LUT, DSP, BRAM, Fmax). No FPGA wall-clock or power measurement exists yet; the procedure is in HARDWARE_TODO.md, and wall-time speedup would be the

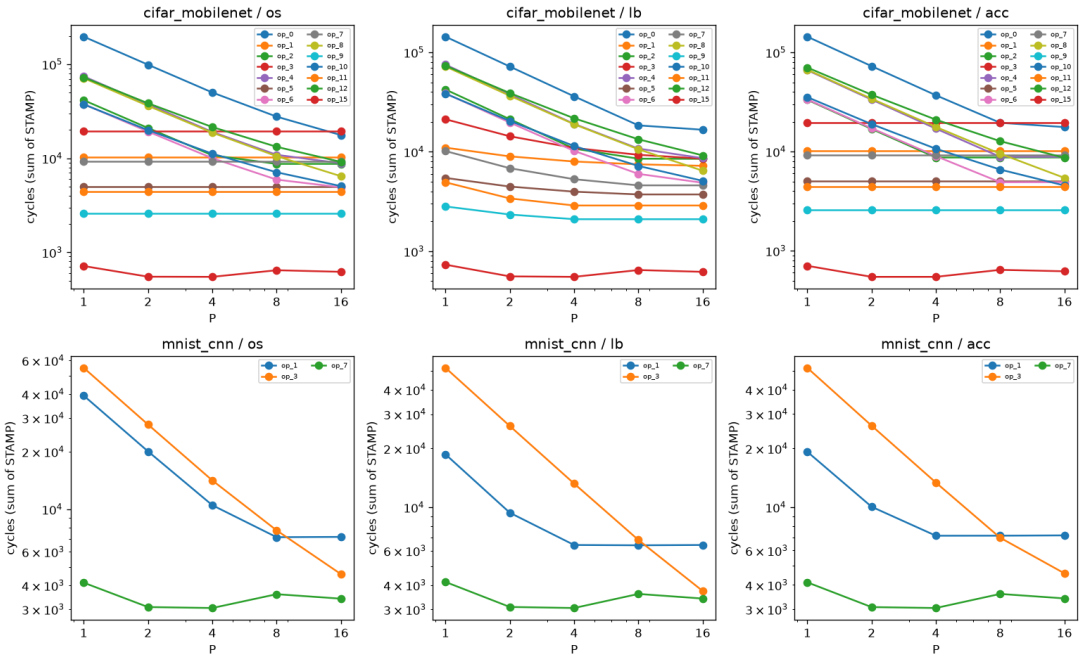


Fig. 2. Predicted per-command stamp cycles versus P (model at tag m1; os is the built architecture, lb and acc are modelled only). Measured overlay: [figure from results/m6_pred_vs_meas.csv after M6].

TABLE III
Synthesis (seed one for resources, Fmax median over seeds).

Design	LUT4	MULT	BRAM	Fmax MHz
PicoRV32	2,896	4	0	78.2
Accelerator, $P = 8$	?	?	?	?
SoC, no accelerator	3,224	4	128	63.1
SoC with accelerator	?	?	?	?

cycle speedup scaled by the Fmax ratio of the SoC with and without the accelerator. The memory model is zero-wait on-chip RAM; an external memory would change the port-bound assumption. The accelerator covers convolution, depthwise (multiplier one) and fully connected layers only; pooling and layout ops stay in software and cap MNIST by Amdahl's law. The model has biases documented before m1, for example one extra setup cycle per pixel where a layer is requantizer bound. Only two networks are evaluated. The repository has no remote, so the m1 hash has not been timestamped externally; this is a manual step.

VIII. Conclusion

We built a small int8 accelerator and its backend and score an analytic model that was frozen before the RTL. The outcomes of the pre-registered claims are those in the tables above; they update from results/ through experiments/make_macros.py.

TABLE IV
Per-command cycles at $P = 8$ (pred. from tag m1; red ? = not yet measured).

net	op	kind	K	pred.	meas.
mnist	1	conv	3x3	7,144	?
mnist	3	conv	3x3	7,756	?
mnist	7	fc	1x1	3,596	?
cifar	0	conv	3x3	27,724	?
cifar	1	dw	3x3	10,236	?
cifar	2	conv	1x1	8,712	?
cifar	3	dw	3x3	19,436	?
cifar	4	conv	1x1	10,888	?
cifar	5	dw	3x3	5,012	?
cifar	6	conv	1x1	5,928	?
cifar	7	dw	3x3	9,204	?
cifar	8	conv	1x1	10,536	?
cifar	9	dw	3x3	2,580	?
cifar	10	conv	1x1	7,096	?
cifar	11	dw	3x3	4,436	?
cifar	12	conv	1x1	13,240	?
cifar	15	fc	1x1	644	?

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